

CHE 201 Fall 2019 Final Exam

Paul Diaz

TOTAL POINTS

97.5 / 100

QUESTION 1

1 15 pts

1.1 a 12 / 12

✓ - 0 pts Correct

- 2 pts first answer is incorrect
- 2 pts second answer incorrect
- 2 pts third answer is incorrect
- 2 pts fourth answer is incorrect
- 2 pts fifth answer is incorrect
- 2 pts sixth answer is incorrect

1.2 b 3 / 3

✓ - 0 pts Correct

- 3 pts incorrect
- 2 pts explanation is wrong

QUESTION 2

2 10 pts

2.1 a 5 / 5

✓ - 0 pts Correct

- 2 pts formula is wrong
- 5 pts incorrect
- 2 pts temperature is not converted to K

2.2 b 5 / 5

✓ - 0 pts Correct

- 5 pts incorrect

QUESTION 3

10 pts

3.1 a 2 / 2

✓ - 0 pts Correct

- 2 pts incorrect

- 1 pts negative sign missing

3.2 b 1.5 / 2

✓ - 0 pts Correct

- 2 pts incorrect

- 0.5 Point adjustment

☞ should be p constant in the second one

3.3 C 6 / 6

✓ - 0 pts Correct

- 1 pts calculation error
- 5 pts Incorrect or no values
- 6 pts Incorrect or no answer
- 2 pts Did not multiply by mass
- 2 pts Wrong value for u
- 3 pts Did not show the calculations for the interpolate values
- 2 pts T should be in Kelvin

QUESTION 4

32 pts

4.1 a 2 / 2

✓ - 0 pts Correct

- 1 pts boundary is wrong

4.2 b 3 / 3

✓ - 0 pts Correct

- 1 pts closed system missing
- 1 pts constant pressure missing
- 1 pts $\Delta KE = \Delta PE = 0$ missing

4.3 C 4 / 4

✓ - 0 pts Correct

- 2 pts Wrong simplified energy balance
- 2 pts Wrong simplified entropy balance

- **1 pts** Need to simplify the entropy balance

4.4 d 8 / 8

✓ - **0 pts** Correct

- **1 pts** unit conversion for work
- **2 pts** v_2 is wrong
- **1 pts** P is wrong
- **1 pts** Work is not multiplied with mass
- **4 pts** Calculation of Q is wrong
- **4 pts** Calculation of W is wrong
- **8 pts** Wrong answer
- **2 pts** u_2 is wrong
- **1 pts** U_1 is wrong
- **1 pts** v_1 is wrong
- **1 pts** Calculation error
- **2 pts** States 1 and 2 are inverted

4.5 e 5 / 5

✓ - **0 pts** Correct

- **1 pts** need to multiply by mass
- **5 pts** Wrong or no answer
- **1 pts** Should use Table A-2 or interpolate Table A-3
- **2 pts** s_2 is wrong
- **2 pts** s_1 is wrong
- **1 pts** Calculation error

4.6 f 4 / 5

- **0 pts** Correct

✓ - **1 pts** T is wrong should be boundary T

- **2 pts** Did not state if process is reversible, irreversible or impossible
- **3 pts** Wrong solution
- **1 pts** Process is impossible
- **1 pts** Calculation or equation error
- **5 pts** No answer

4.7 g 5 / 5

✓ - **0 pts** Correct

- **1 pts** x axis not labeled
- **2 pts** T - s graph is wrong or absent
- **1 pts** v values are wrong or absent

- **1 pts** s value is wrong or absent

- **2 pts** P - v graphs is wrong

- **1 pts** y axis not labeled

- **1 pts** Wrong or missing units

QUESTION 5

5 33 pts

5.1 a 4 / 4

✓ - **0 pts** Correct

- **4 pts** Incorrect
- **1 pts** Did not represent Work
- **1 pts** Wrong system boundary
- **1 pts** Missing one outlet

5.2 b 4 / 5

- **0 pts** Correct

✓ - **1 pts** steady state missing

- **2 pts** control volume missing
- **1 pts** $\Delta PE = 0$ missing
- **1 pts** Cannot assume $\Delta KE = 0$

5.3 c 6 / 6

✓ - **0 pts** Correct

- **2 pts** entropy balance is wrong
- **2 pts** energy balance is wrong
- **2 pts** Q , W , m is not written in rate format
- **1 pts** Need to simplify entropy balance

5.4 d 10 / 10

✓ - **0 pts** Correct

- **1 pts** calculation error
- **1.5 pts** unit conversion error
- **2 pts** h_1 is wrong
- **2 pts** h_2 is wrong
- **2 pts** h_3 is wrong
- **8 pts** Incomplete answer
- **1 pts** Answer should be in MW
- **1 pts** m_1 is wrong
- **2 pts** m_2 and m_3 are wrong
- **1 pts** incomplete calculation

5.5 e 8 / 8

✓ - 0 pts Correct

- 8 pts incorrect or no answer
- 1 pts calculation error
- 2 pts s2 is wrong or absent
- 2 pts s3 is wrong or absent
- 2 pts T is wrong
- 4 pts Incorrect equation or no equation
- 2 pts m2 and m3 are wrong or absent
- 1 pts unit conversion error
- 2 pts s1 is wrong

Total points: 100

Please fill in your answers in the space supplied

Show all of your calculations, conversions and work to receive full credit.

QUESTION 1 (15 pts)

True or False (2 pts each)

True The upstream and downstream enthalpies of flow through an insulated valve is equal. $\rightarrow$ Throttle device $\Rightarrow$ same enthalpies

False Entropy, S, is path-dependent. $\rightarrow$ state function (indep.)

False Gibbs energy is an extensive property. $G = H + TS$
 $\leftarrow$ ext. $\leftarrow$ ext. $\leftarrow$ ext.

False A closed system can experience an increase in entropy only when there is heat transfer from the system.

True The change in Gibbs free energy (dg) is equal to zero for a phase change from a saturated liquid to a saturated vapor.

True The second Carnot corollary states that reversible power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

True for reversible only
 $\eta_{rev} = \eta_{max} = \frac{W_{net}}{Q_H} = \frac{T_H - T_C}{T_H}$

Fill in the blank

(3 pts) A gas goes from State A to State B in a reversible, adiabatic process. It then goes from State B back to State A by a different pathway that is irreversible and not adiabatic. The entropy change, (ΔS) for the irreversible pathway is

equal to (greater than/equal to/ less than) zero. Explain.

$A \rightarrow B$ (rev) : $\Delta S = \left(\frac{\delta Q}{T}\right)_B + 0$ b/c adiabatic
 $\rightarrow \Delta S = 0, S_B = S_A$

$B \rightarrow A$ (irrev) : $\Delta S = \left(\frac{\delta Q}{T}\right)_B + \sigma_{>0}$

Since entropy is a state function and $A \rightarrow B$ is Adiabatic/reversible, The change in entropy is zero b/c $S_A = S_B$

QUESTION 2 (10 pts) ^{→ power cycle}

- a) (5 pts) A steam engine operates with a theoretical maximum efficiency of %50 and a minimum temperature of 150°C. Calculate temperature of the engine to maintain maximum efficiency.

$$\eta_{\max} = \frac{w_{\text{cycle}}}{T_H} = \frac{T_H - T_C}{T_H}, \quad T_C = 150^\circ\text{C} + 273.15\text{K}$$

$$T_C = 423.15\text{K}$$

$$0.50 = \frac{T_H - 423.15\text{K}}{T_H}$$

$$0.5T_H = T_H - 423.15\text{K}$$

$$-0.5T_H = -423.15\text{K}$$

$$T_H = 846.3\text{K}$$

- b) (5 pts) The Reynolds number, Re , is a dimensionless number that measures the ratio of inertial forces to viscous forces in fluid flow;

$$Re = \frac{\rho V L}{\mu}$$

In this equation, ρ is the density (kg/m^3), V is the velocity (m/s), L is a characteristic length (m), and μ is the viscosity. What are the dimensions of μ ?

Show all your work for full credit.

$$\rho [=] \text{kg/m}^3$$

$$V [=] \text{m/s}$$

$$L [=] \text{m}$$

$$\mu [=] ?$$

$$(no\ dimensions) = \frac{\rho V L}{\mu}$$

$$\mu = \rho V L$$

$$\mu [=] \frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{m}}{\text{s}} \cdot \text{m}$$

$$\mu = \text{kg/ms}$$

or

$$\mu = \text{kg m}^{-1} \text{s}^{-1}$$

$$dz = Mdx + Ndy$$

QUESTION 3 (10 pts)

- a) (2 pts) Write the specific Maxwell relation that corresponds to the thermodynamic function:

u(s, v): $\left(\frac{\partial T}{\partial v}\right)_s = -\left(\frac{\partial P}{\partial s}\right)_v$

$$dw = Tds - Pdv$$

Maxwell relation

$$\left(\frac{\partial M}{\partial Y}\right)_X = \left(\frac{\partial N}{\partial X}\right)_Y$$

- b) (2 pts) Write the two, second mixed partial derivatives for the exact differential:

dh = Tds + vdP : $\left(\frac{\partial T}{\partial P}\right)_s, \left(\frac{\partial v}{\partial s}\right)_P$

exact differential

$$\left(\frac{\partial M}{\partial Y}\right)_X = \left(\frac{\partial N}{\partial X}\right)_Y$$

- b) (6 pts) A 3 kg mass of saturated liquid water has a pressure of 10.0 bar. What is the Helmholtz energy, in kJ?

Show all your work for full credit

Helmholtz free energy $F = U - TS$

$$T = 179.9^\circ\text{C} + 273.15$$

$$T = 453.05\text{K}$$

Specific Helmholtz $\psi = u - Ts$

$$u_f(10\text{ bar}, T_{\text{sat}}^{179.9^\circ}) = 761.68\text{ kJ/kg}$$

$$s_f(10\text{ bar}, T_{\text{sat}}) = 2.1387\text{ kJ/kg K}$$

$$\psi = 761.68\text{ kJ/kg} - (453.05\text{ K})(2.1387\text{ kJ/kg K})$$

$$\psi = -207.26\text{ kJ/kg}$$

$$F = \psi \cdot m = -207.26 \cdot 3\text{ kg}$$

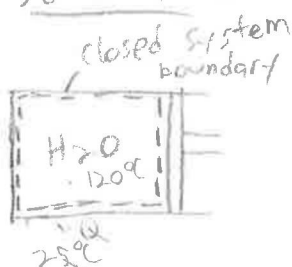
$$F = -621.77\text{ kJ}$$

QUESTION 4: (32 pts)

3 kg of water, initially a saturated liquid at 120°C is contained piston-cylinder device. The water undergoes a process. Heat is transferred between the piston-cylinder and a 25°C room until the water becomes a state where $x=0.8$. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (3 pts) State the engineering model
- (4 pts) Write the simplified energy balance and simplified entropy balance.
- (8 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.
- (5 pts) Determine the change in entropy of the water during the process, **ΔS , in kJ/K**
- (5 pts) Determine the entropy generated during the process, **σ , in kJ/K**. State if this process is reversible, irreversible or impossible.
- (5 pts) Carefully sketch the process on P-v and T-s diagram.

a) schematic



b) Engineering Model

- closed system
- Piston moves freely w/ constant P
- $\Delta KE = \Delta PE = 0$

c) Energy balance

$$\Delta E = \cancel{\Delta KE} + \cancel{\Delta PE} + \Delta U = Q - W$$

$$\Delta U = Q - W$$

Entropy balance

$$S_2 - S_1 = \int_1^2 \left(\frac{\delta Q}{T} \right)_b + \sigma$$

$$S_2 - S_1 = \frac{Q_{\text{closed system}}}{T_b} + \sigma$$

$$T_b = 120^\circ\text{C}$$

+w → done by syst/expans

d) $W, Q = ?$

$$W = \int_1^2 P dV = P \int_1^2 dV = P \Delta V = P(V_2 - V_1)$$

$$V_1 = v_f(120^\circ\text{C}) \cdot m = (1.0603 \cdot 10^{-3} \text{ m}^3/\text{kg}) \cdot 3 \text{ kg} = 3.18 \cdot 10^{-3} \text{ m}^3$$

$$V_2 = (v_f(120^\circ\text{C}) + x_r(v_g - v_f)) \cdot m = 0.714 \cdot 3 = 2.141 \text{ m}^3$$

$$v_2 = 1.0603 \cdot 10^{-3} + 0.8(0.8919 - 1.0603 \cdot 10^{-3}) = 0.714 \text{ m}^3/\text{kg}$$

$$W = P(V_2 - V_1) = 1.985 \text{ bar} \cdot (2.138 \text{ m}^3) \cdot \frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \cdot \frac{1 \text{ kJ}}{1000 \text{ Nm}}$$

$$W = 424.55 \text{ kJ (expansion)}$$

$$\Delta U = m(u_2 - u_1) = 3 \text{ kg} (1620.64 \text{ kJ/kg}) = 4861.92 \text{ kJ}$$

$$u_1 = u_f(120^\circ\text{C}) = 503.5 \frac{\text{kJ}}{\text{kg}}; u_2 = 503.5 + 0.8(2529.3 - 503.5)$$

$$u_2 = 2124.14 \text{ kJ/kg}$$

$$U = Q - W \Rightarrow Q = U + W = 4861.92 + 424.55$$

$$Q = 5286.47 \text{ kJ}$$

e) $\Delta S = ?$

$$\Delta S = m(s_2 - s_1) = 3 \text{ kg} (4.4816 \text{ kJ/kg K}) ; \Delta S = 13.4448 \frac{\text{kJ}}{\text{K}}$$

$$s_1 = s_f(120^\circ\text{C}) = 1.5276 \text{ kJ/kg K}$$

$$s_2 = s_f + x_r(s_g - s_f) = 6.0092$$

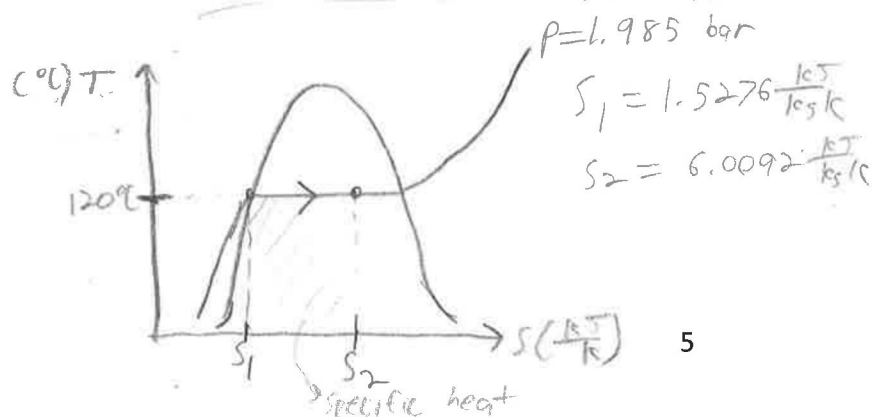
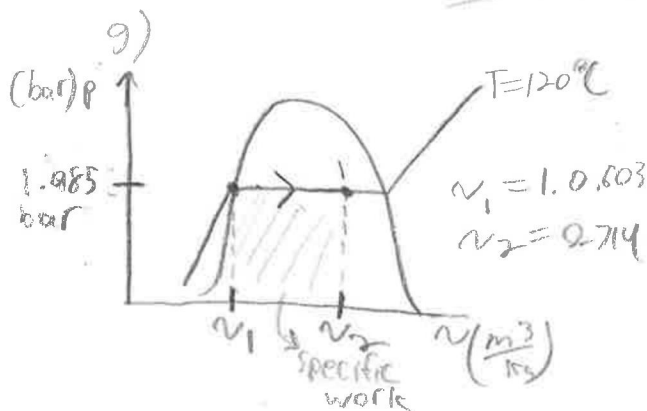
$60.8 \rightarrow 7.1296$

f) $\sigma = ?$

$$\Delta S = 13.4448 \frac{\text{kJ}}{\text{K}} = \frac{5286.47 \text{ kJ}}{393.15 \text{ K}} + \sigma$$

$$\sigma = 13.4448 - 13.797 \frac{\text{kJ}}{\text{K}}$$

$$(\sigma = -0.3526 \frac{\text{kJ}}{\text{K}}) \Rightarrow \text{Impossible b/c } \sigma \leq 0$$



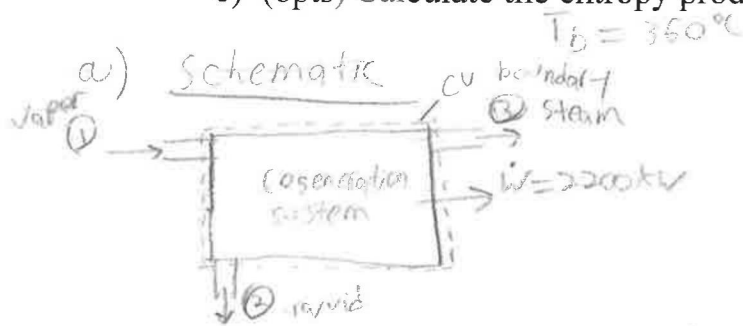
QUESTION 5: (33 pts)

Develop power?

$$\dot{W} = + 2.2 \cdot 10^6 \frac{\text{J}}{\text{s}} \cdot \frac{1 \text{ kW}}{1000 \text{ J/s}} = 2200 \text{ kW}$$

A cogeneration system operates at steady state. Water vapor enters the system at 20 bar, 360°C, velocity of 50 m/s and mass flow rate of 1.5 kg/s. Power is developed by the system at a rate of 2.2 MW. There are two outlets for the system: process steam and hot water. Process steam leaves as saturated vapor at 1 bar, 100 m/s, and hot water leaves as saturated liquid at 1 bar, 100 m/s. Mass flow rates of process steam and hot water are equal. Neglect any changes in potential energy.

- (4pts) Draw a schematic of the system and label the system boundary.
- (5pts) State the engineering model.
- (6pts) Write simplified balances for mass, energy and entropy.
- (10pts) Calculate the rate of heat transfer, in MW, between the system and its surroundings.
- (8pts) Calculate the entropy production for the process, in kJ/K-s.



b) Engineering Model

- control volume at steady state
- one inlet / 2 outlets
- $\Delta PE = 0$

c) Mass balance

$$\frac{d m_{cv}}{dt} = 0 = \dot{m}_1 - \dot{m}_2 - \dot{m}_3 \quad \text{so} \quad \dot{m}_1 = \dot{m}_2 + \dot{m}_3$$

$$\dot{m}_1 = 1.5 \text{ kg/s}; \quad \dot{m}_2 = \dot{m}_3 = 0.75 \text{ kg/s}$$

Energy balance

$$\frac{d E_{cv}}{dt} = 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left(h_1 + \frac{V_1^2}{2} + g z_1 \right) - \dot{m}_2 \left(h_2 + \frac{V_2^2}{2} + g z_2 \right) - \dot{m}_3 \left(h_3 + \frac{V_3^2}{2} + g z_3 \right)$$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left(h_1 + \frac{V_1^2}{2} \right) - \dot{m}_2 \left(h_2 + \frac{V_2^2}{2} \right) - \dot{m}_3 \left(h_3 + \frac{V_3^2}{2} \right)$$

Entropy balance

$$\frac{d S_{cv}}{dt} = 0 = \frac{\delta Q}{T_b} + \dot{m}_1 s_1 - \dot{m}_2 s_2 - \dot{m}_3 s_3 + \dot{S}_{cv}$$

$$0 = \frac{\dot{Q}_{cv}}{T_b} + \dot{m}_1 s_1 - \dot{m}_2 s_2 - \dot{m}_3 s_3 + \dot{S}_{cv}$$

$$T_2 = T_3 = 99.63^\circ\text{C}$$

d) Find $\dot{Q}_{cv}$

$$\dot{w}_{cv} = +2200 \frac{\text{kJ}}{\text{s}}; \quad h_1 = h(20 \text{ bar}, 360^\circ\text{C}) = 3159.3 \frac{\text{kJ}}{\text{kg}}$$

$$h_2 = h_f(1 \text{ bar}) = 417.46 \frac{\text{kJ}}{\text{kg}}; \quad h_3 = h_g(1 \text{ bar}) = 2675.5 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{V_1^2}{2} = \frac{(50 \text{ m/s})^2}{2} = 1250 \text{ m}^2/\text{s}^2 = 1250 \text{ J/kg} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = 1.25 \frac{\text{kJ}}{\text{kg}}$$

$$\frac{V_2^2}{2} = \frac{V_3^2}{2} = \frac{(100 \text{ m/s})^2}{2} = 5000 \text{ m}^2/\text{s}^2 = 5000 \text{ J/kg} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = 5 \frac{\text{kJ}}{\text{kg}}$$

$$0 = \dot{Q}_{cv} - 2200 \text{ kW} + 1.5 \text{ kg/s} (3159.3 + 1.25 \frac{\text{kJ}}{\text{kg}}) - 0.75 \text{ kg/s} (417.46 + 5 \frac{\text{kJ}}{\text{kg}}) - 0.75 \text{ kg/s} (2675.5 + 5 \frac{\text{kJ}}{\text{kg}})$$

$$0 = \dot{Q}_{cv} - 2200 + 4740.825 - 316.845 - 2010.375 \frac{\text{kJ}}{\text{s}}$$

$$0 = \dot{Q}_{cv} + 213.605 \text{ kW} \Rightarrow \dot{Q}_{cv} = -213.605 \frac{\text{kJ}}{\text{s}}$$

$$\dot{Q}_{cv} = -213.605 \cdot 10^3 \frac{\text{J}}{\text{s}} \cdot \frac{1 \text{ MJ}}{10^6 \text{ J}} = -0.213605 \text{ MW}$$

$$\boxed{\dot{Q}_{cv} = -0.213605 \text{ MW}}$$

e) Find $\dot{S}_{cv}$

$$0 = \frac{(-213.605 \text{ kJ/s})}{633.15 \text{ K}} + (1.5 \frac{\text{kg}}{\text{s}})(6.9917 \frac{\text{kJ}}{\text{kg K}}) - 0.75(1.3028) - 0.75(7.3594) + \dot{S}_{cv}$$

$$0 = -0.337 \frac{\text{kJ}}{\text{K s}} + 10.486 \frac{\text{kJ}}{\text{K s}} - 0.977 \frac{\text{kJ}}{\text{K s}} - 5.519 \frac{\text{kJ}}{\text{K s}} + \dot{S}_{cv}$$

$$0 = \dot{S}_{cv} + 3.654 \frac{\text{kJ}}{\text{K s}}$$

$$\boxed{\dot{S}_{cv} = -3.654 \frac{\text{kJ}}{\text{K s}}}$$

